

Thesis Defense:
Invariant measures for the stochastic heat equation with
no drift term

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The stochastic heat equation

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2}\Delta_x u(t, x) = b(x, u(t, x))\dot{W}(t, x), & t \geq 0, x \in \mathbb{R}^d \\ u(0, \cdot) = \mu(\cdot) \end{cases}$$

- The Stochastic Heat Equation is interpreted as follows:

$$u(t, x) = \int_{\mathbb{R}^d} G(t, x - y)\mu(dy) + \int_0^t \int_{\mathbb{R}^d} G(t - s, x - y)b(u(s, y))W(ds, dy)$$

where $G(t, x)$ is the heat kernel

$$G(t, x) = (2\pi t)^{-d/2} \exp\left(-\frac{|x|^2}{2t}\right)$$

- A processes, $\{u(t, x) : t \geq 0, x \in \mathbb{R}^d\}$, that satisfies this relation is referred to as the *mild solution*.

The noise term $\dot{W}$

- $W = \{W(\phi) : \phi \in \mathcal{S}(\mathbb{R}^d)\}$ is a centered Gaussian process uniquely defined through the following covariance functional:

$$J(\phi, \psi) := \mathbb{E}[W(\phi)W(\psi)] = \int_0^\infty ds \int_{\mathbb{R}^d} \Gamma(dx)(\phi(s, \cdot) * \tilde{\psi}(s, \cdot))(x),$$

where $\tilde{\psi}(s, x) = \psi(s, -x)$. We have the following nomenclature.

- *Space-time white noise*: $\Gamma(dx) = \delta_0(dx)$.
- *Time white and spatially colored*: $\Gamma(dx) = f(x)dx$ where f is a non-negative and non-negative definite function referred to as the *Correlation function* and its Fourier transform,

$$\hat{f}(\xi) := \mathcal{F}f(\xi) = \int_{\mathbb{R}^d} \exp(-ix \cdot \xi) f(x)dx,$$

is referred to as the *spectral density*.

Proposition (DaPrato and Zabczyk [7, Propsition 4.19])

A space-time white noise is the distributional derivative of a Brownian sheet, $B(t, x)$.

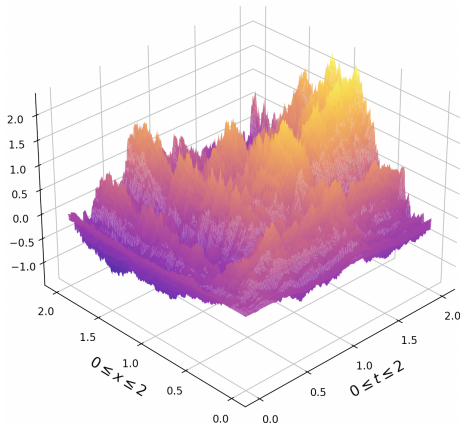


Figure: Brownian Sheet Simulation (Walsh [8]):

$$B(t, x) \sim B(t - \delta_t, x) + B(t, x - \delta_x) - B(t - \delta_t, x - \delta_x) + \text{normal}(0, \delta_t \delta_x)$$

$$B(t, 0) = B(0, x) = 0 \text{ for all } t, x > 0.$$

Existence and uniqueness with space-time white noise

- A necessary condition for global existence, *Dalang's condition* [6], is the following:

$$\int_0^t ds \int_{\mathbb{R}^d} dy G^2(s, y) < \infty, \quad \forall t > 0.$$

- Direct calculation yields that,

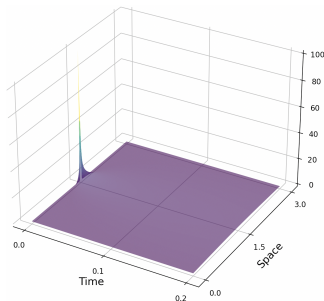
$$\int_0^t ds \int_{\mathbb{R}^d} dy G^2(s, y) = \frac{\Gamma(d/2)}{2(2\pi)^d} \int_0^t \frac{1}{s^{d/2}} ds,$$

which forces,

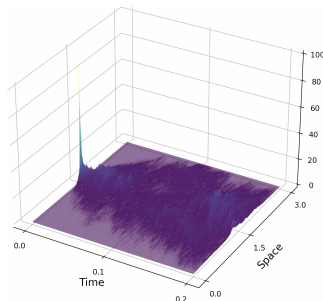
$$d/2 < 1 \quad \Longleftrightarrow \quad d < 2.$$

- Thus a global space-time white noise solution may only exist when the spatial dimension is 1.

Large Peaks: $u_t - \frac{1}{2}u_{xx} = \lambda u \dot{W}$ with $u_0(x) = \delta_0(x)$
 driven by a space-time white noise



$\lambda = 0$

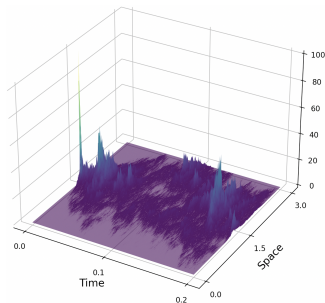


$\lambda = 1$

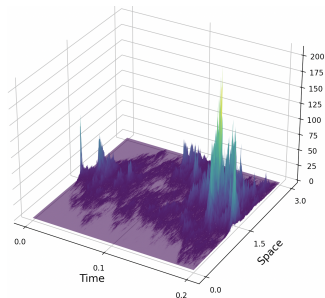
- Simulations done by using a finite difference scheme [5].
- We partition (space) $[0, 3]$ into n_x pieces of size $\delta_x = 3/n_x$.
- We partition (time) $[0, .2]$ into n_t pieces of size $\delta_t = .2/n_t$.

$$u(t + \delta_t, x) = u(t, x) + n_x^2 \delta_t (u(t, x + \delta_x) + u(t, x - \delta_x) - 2u(t, x)) \\
+ n_x \lambda u(t, x) \text{normal}(0, \delta_t \delta_x).$$

Intermittency of $u_t - \frac{1}{2}u_{xx} = \lambda u \dot{W}$ driven by space-time white noise



$\lambda = 1.8$



$\lambda = 2$

● Notice the enlargement of the peaks as λ increases. We have the following theorem.

Theorem (Chen and Dalang [1, Theorem 2.4])

The second moments of the stochastic heat equation driven by space-time white noise grow exponentially in time.

Will the second moments ever be bounded in time?

- Several slides below will require the second moments to be bounded in time:

$$\sup_{t \geq 0} \|u(t, x)\|_2 < \infty, \quad \text{where } \|X\|_p = [\mathbb{E}(|X|^p)]^{1/p}.$$

- This is possible, however this will require us to only consider $d \geq 3$, and therefore we must only consider a spatially colored noise.
- Recall from above, that the covariance of a spatially colored noise is given by the following:

$$\begin{aligned} \mathbb{E}[W(\phi)W(\psi)] &= \int_0^\infty ds \int_{\mathbb{R}^d} dx f(x)(\phi(s, \cdot) * \tilde{\psi}(s, \cdot))(x) \\ &= \int_0^\infty ds \int_{\mathbb{R}^{2d}} dx dy \phi(s, x) f(x - y) \psi(s, y) \\ &= (2\pi)^{-d} \int_0^\infty ds \int_{\mathbb{R}^d} d\xi \hat{f}(\xi) \mathcal{F}\phi(s, \xi) \overline{\mathcal{F}\psi(s, \xi)} \end{aligned}$$

Existence and Uniqueness with a Colored Noise

Theorem (Chen and Huang [2, Theorem 1.2])

There exists a unique global mild solution under the following:

- $b : \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}$ such that there exists a $L_b, L_0 > 0$ so for all $u, v \in \mathbb{R}$ and $x \in \mathbb{R}^d$,

$$|b(x, u) - b(x, v)| \leq L_b |u - v| \quad \text{and} \quad |b(x, 0)| \leq L_0.$$

- The initial measure, $\mu(dx)$ (resp. $\varphi(x)dx$), satisfies for some $a > 0$,

$$\int_{\mathbb{R}^d} e^{-a|x|^2} |\mu|(dx) < \infty \quad \left(\text{resp.} \quad \int_{\mathbb{R}^d} e^{-a|x|^2} |\varphi(x)| dx < \infty \right),$$

where $|\mu| = \mu^+ + \mu^-$ is the *Hahn decomposition*.

- The spectral density, $\hat{f}$, satisfies *Dalang's condition* for some $\beta > 0$:

$$\Upsilon(\beta) := \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{\beta + |\xi|^2} < \infty.$$

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